



BYTEDANCE SEED



# VAPO: Value-Model-Based RL for Advanced Reasoning

60.4 on AIME 2024 — Surpassing DAPO by 10+ Points



Built on Qwen2.5-32B with No SFT Data



# Why Value Models? Finer Credit Assignment for Long CoT

## Value-Model-Free (GRPO, DAPO)

Trajectory reward  $r(\tau)$  assigned uniformly  
to all tokens:  $A(t) = r(\tau) - \text{baseline}$

### Limitations for complex reasoning:

- Same credit for correct and incorrect intermediate steps
- High variance from Monte Carlo returns
- Poor sample efficiency on long trajectories

## Value-Model-Based (VAPO)

Token-level value estimates  $V(s_t)$  enable  
precise per-token advantage:  $A(t) = \delta_t + \gamma \lambda \delta_{t+1} + \dots$



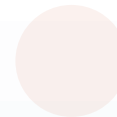
### Precise Credit Assignment

Trace each action's impact on returns;  
critical when subtle errors cause catastrophic failures



### Variance Reduction

Lower-variance per-token estimates  
compared to Monte Carlo methods



### Generalization

Well-trained value model generalizes  
across samples, improving sample efficiency

# Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks



## Value Model Bias

Long trajectories + bootstrapping

Value estimates accumulate error across thousands of tokens.

Bootstrapped targets drift, causing severe bias in early training.

CRITICAL



## Heterogeneous Lengths

Different bias-variance tradeoffs

Short and long responses need different  $\lambda$  in GAE.

Fixed  $\lambda$  fails across the full spectrum of reasoning depths.

NOVEL



## Reward Signal Sparsity

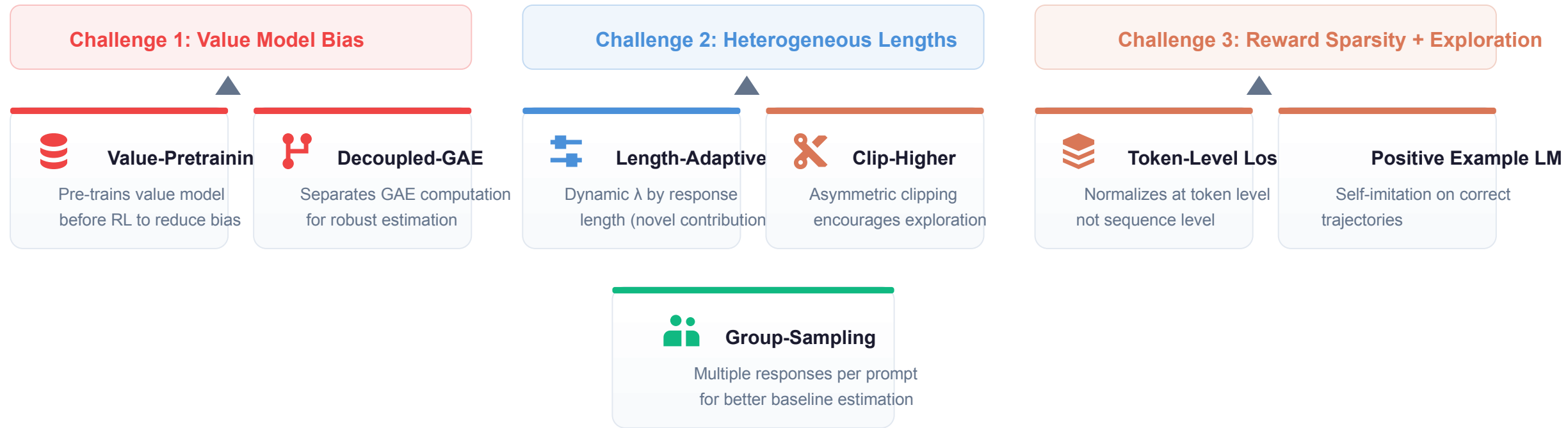
Only final token receives reward

In long CoT, only the `<eos>` token gets a scalar reward.

Credit must propagate backward through thousands of steps.

HARD

# VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL



## Integrated VAPO Pipeline

Qwen2.5-32B (no SFT) → Value-Pretraining → Decoupled-GAE + Length-Adaptive GAE → PPO with Clip-Higher + Token-Level Loss + Positive Example LM Loss + Group-Sampling

**Result: 60.4 AIME 2024 | 5,000 steps | Zero training crashes**

# Length-Adaptive GAE: Dynamic $\lambda$ Adjustment for Variable Response Lengths

## GAE Advantage Estimation

$$A^{\text{GAE}}(t) = \sum (\gamma \lambda)^l \cdot \delta_{t+l}$$

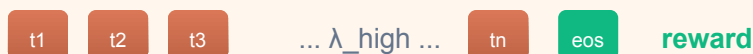
### Short Response

Lower  $\lambda \rightarrow$  lower variance (fewer bootstrap steps)



### Long Response

Higher  $\lambda \rightarrow$  lower bias (more Monte Carlo, less bootstrap)



## Why It Matters

- Fixed  $\lambda$  fails across heterogeneous lengths
- Short responses: bootstrap bias dominates
- Long responses: Monte Carlo variance dominates
  - Length-adaptive  $\lambda$  optimizes the tradeoff per sample
- Ablation: removing it drops score from 60  $\rightarrow$  45

## Implementation

$\lambda(\text{response\_length}) = f(\text{len})$  where:

- $\text{len} < L_{\text{short}} \rightarrow \lambda = \lambda_{\text{min}}$  (low variance)
- $\text{len} > L_{\text{long}} \rightarrow \lambda = \lambda_{\text{max}}$  (low bias)
- Linear interpolation in between

**Contribution: +15 points on AIME 2024**

# VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



Key Facts

Same base:  
Qwen2.5-32B

No SFT data

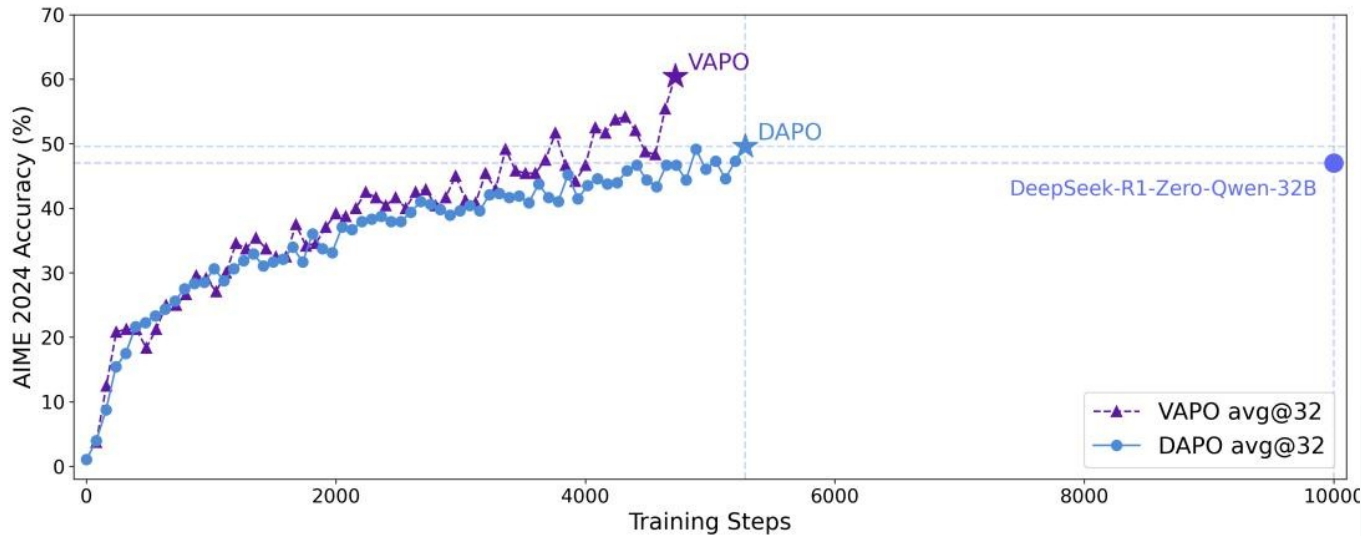
Metric:  
AIME24 avg@32



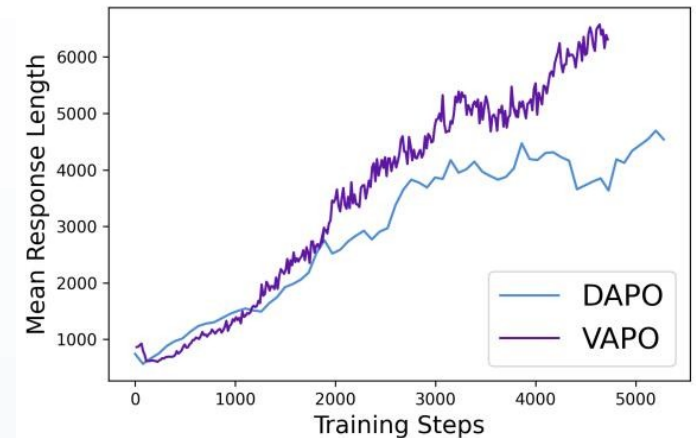
See training curve (figure\_2.jpg): VAPO consistently outperforms DAPO throughout training, reaching 60% at ~4,500 steps vs DAPO plateauing near 50% at ~5,500 steps

# VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO

Training Curve: Accuracy vs Steps (figure\_2.jpg)



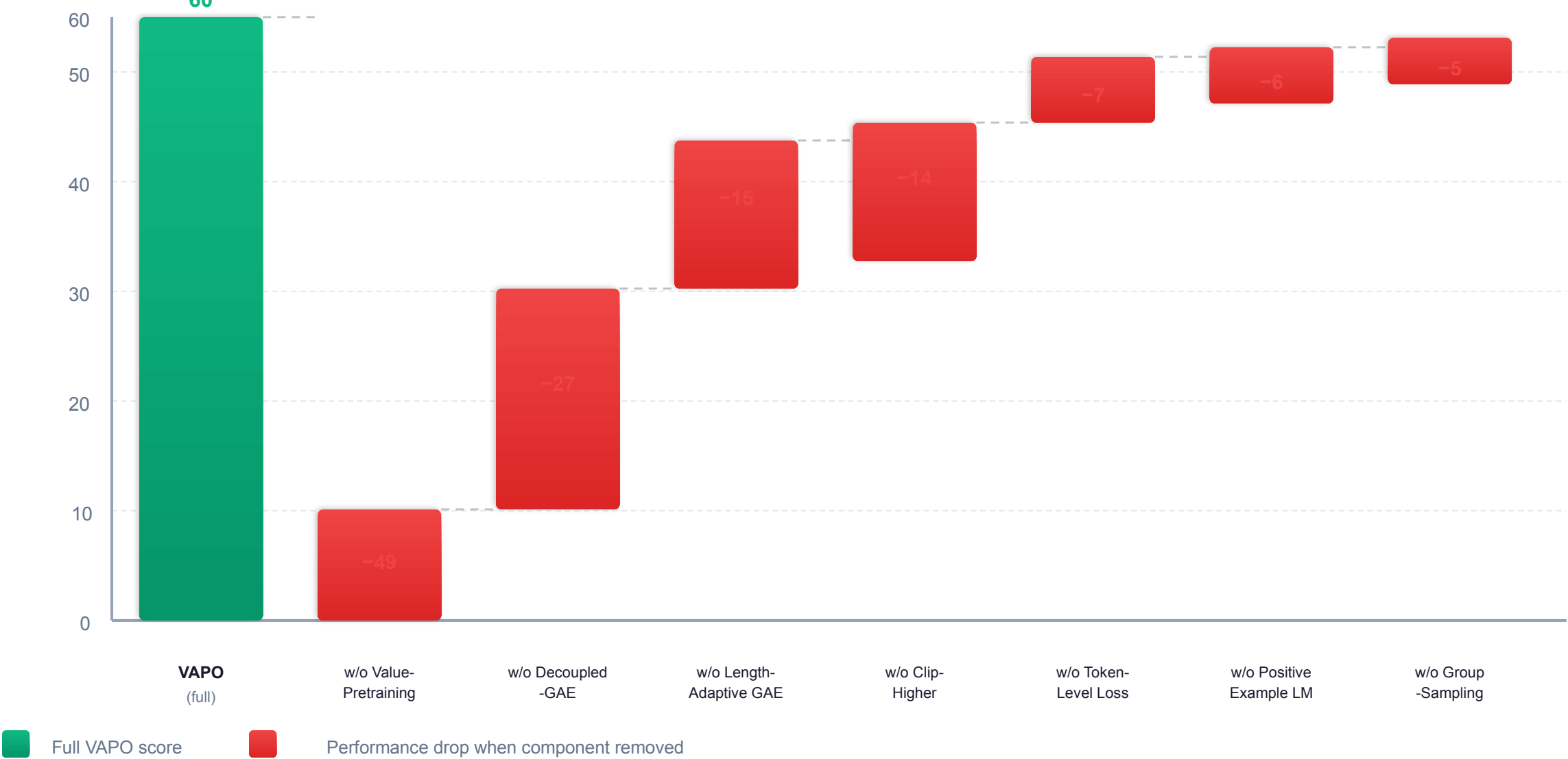
Response Length (figure\_3.jpg)



## Key Efficiency Findings

- VAPO reaches 60% accuracy at ~4,500 steps — DAPO plateaus near 50% at ~5,500 steps
- DeepSeek-R1-Zero-Qwen-32B reaches 47% only after ~10,000 steps (2× training budget)
- VAPO generates longer reasoning chains (~6,000 tokens vs DAPO ~4,500) training elaborates reasoning multiple runs

# Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points





# Value-Model-Based RL Is Back

— With the Right Engineering

1

## Surpasses Value-Free Methods

VAPO (60) beats DAPO (50) and GRPO when core challenges are systematically addressed.

2

## Length-Adaptive GAE Is Critical

Novel dynamic  $\lambda$  adjustment contributes 15 points — essential for heterogeneous lengths.

3

## Value Pretraining = 49 Points

Removing it drops score from 60 to 11. Without it, value-model-based RL barely works.

4

## Highly Stable Training

Zero training crashes across multiple independent runs — rare for value-based RL.

5

## Superior Training Efficiency

Reaches SOTA in ~5,000 steps vs DAPO's ~5,500 for a lower score — 10× faster than DeepSeek.

6

## Future Implications

Value-model-based RL should be revisited for all long-CoT reasoning model development.

arXiv:2504.05118 | ByteDance Seed